

Lecture 10

The Harmonic Oscillator

Art: *I think I see it, Lenny. The whole picture is slowly coming into focus. Minus One, General Uncertainty, entangled pairs, the Hamiltonian—even the degenerates. What's next?*

Lenny: *Oscillations, Art. Vibrations. You're a fiddler—play us a last tune tonight. Something with good vibes.*

Of all the ingredients that go into building a quantum description of the world, two stand out as especially fundamental. The spin, or qubit, of course is one of them. In classical logic, everything can be built out of yes-no questions. Similarly, in quantum mechanics, every logical question boils down to a question about qubits. We spent a lot of time in earlier Lectures learning about qubits. In this lecture, we'll learn about the second basic ingredient of quantum mechanics—the harmonic oscillator.

The harmonic oscillator isn't a particular object like a hydrogen atom or a quark. It's really a mathematical framework for understanding a huge number of phenomena. This concept of the harmonic oscillator also exists in classical physics, but it really comes to the fore in quantum theory.

One example of a harmonic oscillator is a particle moving under a linear restoring force; for example, the iconic weight on the end of a spring. An idealized spring satisfies Hooke's law: the force on the displaced mass is